

CMPT 727 Final Project Research Playbook

From Proposal to a Conference-Style Final Report

Working research direction

When Does Logic Help?

Probabilistic Relational Priors for Low-Label Graph Neural Networks

Waleed Ahmed

Student ID: 301663286

CMPT 727: Statistical Machine Learning

Spring 2026

Purpose. This document is a research workflow and planning guide, not a completed project. It starts from the submitted proposal (GCN + homophily/logic regularization on citation networks) and develops a stronger, course-aligned experimental study involving graph smoothness, an explicit pairwise MRF, approximate inference, controlled synthetic graphs, and careful statistical evaluation.

Contents

1	What the Final Project Should Become	4
1.1	Recommended final title	4
2	Conference Template Recommendation	4
2.1	Primary recommendation: ICLR 2026 style	4
2.2	Alternative: NeurIPS 2026 style	5
2.3	Do not optimize for the template too early	5
3	Research Questions and Hypotheses	5
3.1	RQ1: Does the relational prior help most when labels are scarce?	5
3.2	RQ2: Does prior usefulness depend on graph label structure?	5
3.3	RQ3: Is an explicit probabilistic prior better motivated than a heuristic smoothness penalty?	5
3.4	RQ4: Can a context-aware prior be more robust than unconditional homophily?	5
3.5	Optional RQ5: What happens to predictive uncertainty?	5
4	Scope: What Is Essential vs. Optional	5
5	Datasets to Use	6
5.1	Dataset A: Cora – primary dataset	6
5.2	Dataset B: CiteSeer – replication dataset	6
5.3	Dataset C: Synthetic stochastic block model – most important extra experiment	6
5.4	Dataset D: Roman-empire – optional stress test	7
5.5	Dataset decision	7
6	Project Setup from Zero	7
6.1	Step 0: Freeze the scientific contract	7
6.2	Step 1: Create a reproducible repository	8
6.3	Step 2: Fix reproducibility	9
7	Model Progression	9
7.1	M0: Feature-only MLP	9
7.2	M1: Standard GCN	9
7.3	M2: GCN + Laplacian/logic regularization – proposal model	10
7.4	M3: Probability-of-rule-violation regularizer	10
7.5	M4: GCN unary potentials + explicit Potts MRF	10
7.6	M4 inference: mean-field variational updates	10
7.7	M5: Similarity-weighted, exception-aware prior	11

8 Data Splits and Low-Label Protocol	11
8.1 Do not rely only on the public Planetoid split	11
8.2 Validation discipline	12
9 Hyperparameter Plan	12
10 Metrics to Compute and Save	12
10.1 Primary predictive metrics	12
10.2 Graph/prior diagnostics	13
10.3 Rule-satisfaction metric	13
11 The Experiment Matrix	13
11.1 Experiment E0: Baseline sanity check	13
11.2 Experiment E1: Reproduce the proposal	13
11.3 Experiment E2: Low-label study – core result	14
11.4 Experiment E3: Prior-strength sensitivity	14
11.5 Experiment E4: Explicit MRF vs. smoothness regularization	14
11.6 Experiment E5: Synthetic homophily sweep – centerpiece experiment	14
11.7 Experiment E6: CiteSeer replication	15
11.8 Experiment E7: Heterophily stress test – optional	15
11.9 Experiment E8: Uncertainty/calibration – optional	15
12 What Results You Should Obtain vs. What You Should Never Chase	15
13 Run-Level Results File: Exact Schema	16
13.1 Save predictions for important runs	16
14 Statistical Reporting	16
15 Figures and Tables to Produce	17
15.1 Main-paper figures	17
15.2 Main-paper tables	17
16 Figure Design Rules	17
17 End-to-End Workflow	18
18 A Practical 14-Day Schedule	18
19 Implementation Checks Before Large Runs	19
19.1 GCN checks	19

19.2 Laplacian loss checks	19
19.3 MRF checks	19
19.4 Synthetic graph checks	19
20 How to Write the Final Conference Paper	20
20.1 Recommended 9-page budget	20
20.2 Introduction structure	20
20.3 Methods structure	20
20.4 Results structure	21
20.5 Discussion/Future Work	21
21 Appendix Plan	21
22 Ablations That Matter	22
23 Common Failure Modes and How to Avoid Them	22
24 Decision Rules When Experiments Disagree	22
25 Minimum Viable Paper vs. Excellent Paper	23
25.1 Minimum viable, still solid	23
25.2 Excellent version	23
25.3 Do not cross into unnecessary complexity	23
26 Your First Working Session: Exact Checklist	24
27 Final Submission Checklist	24
27.1 Scientific completeness	24
27.2 Report quality	24
27.3 Reproducibility	25
28 Recommended Reading: Only What Directly Helps the Project	25
29 The One-Sentence Standard for the Final Paper	25

1 What the Final Project Should Become

Your submitted proposal already has a coherent foundation:

- a GCN baseline for node classification;
- the Cora citation network;
- a homophily assumption: connected papers are likely to have the same topic;
- a differentiable logic/smoothness regularizer;
- low-label experiments designed to test whether the prior helps when supervision is scarce.

The strongest next step is *not* to replace that proposal with a completely different project. Instead, turn it into a deeper research question:

Recommended central question

When, why, and under what graph conditions does a relational prior improve GNN node classification, and when does the same prior become harmful?

That question gives the paper a scientific story rather than a simple “our regularizer improves accuracy” story. It also creates space for positive results, negative results, and mechanistic interpretation.

1.1 Recommended final title

A strong working title is:

When Does Logic Help? Probabilistic Relational Priors for Low-Label Graph Neural Networks

Alternative title if the explicit MRF becomes the main contribution:

From Graph Smoothness to Probabilistic Logic: MRF-Regularized GNNs for Low-Label Node Classification

2 Conference Template Recommendation

2.1 Primary recommendation: ICLR 2026 style

Use the **ICLR 2026 LaTeX template** for the course report. The official ICLR 2026 author guide limits the main submission text to 9 pages and allows 10 pages for camera-ready, while references do not count and appendices may follow the bibliography. This fits the course requirement of 6–10 main pages unusually well. It is also thematically natural for GNNs, representation learning, and probabilistic/neural methods.

Suggested target length

Aim for **8.5–9.5 pages of main paper**, plus references. Put implementation details, full hyperparameter grids, extra tables, and extra plots in an appendix.

Official template source: <https://github.com/ICLR/Master-Template/raw/master/iclr2026.zip>

2.2 Alternative: NeurIPS 2026 style

NeurIPS 2026 also uses a 9-page main-text limit and permits an additional content page for camera-ready. It is a good alternative if you prefer its visual style. For a course report, however, ICLR is slightly simpler because its page budget almost exactly matches the instructor’s 6–10 page requirement.

2.3 Do not optimize for the template too early

Do the research in a normal working notebook/document first. Switch to the conference template when the experiment design is frozen and the main plots exist. Otherwise, page limits encourage premature editing before the scientific story is clear.

3 Research Questions and Hypotheses

Write the project around explicit, falsifiable questions.

3.1 RQ1: Does the relational prior help most when labels are scarce?

H_1 : $\Delta\text{Acc} = \text{Acc}_{\text{prior}} - \text{Acc}_{\text{GCN}}$ increases as the number of labeled nodes decreases.

3.2 RQ2: Does prior usefulness depend on graph label structure?

H_2 : The prior helps on graphs where neighboring labels are informative, and can hurt when the homophily assumption is violated.

3.3 RQ3: Is an explicit probabilistic prior better motivated than a heuristic smoothness penalty?

Compare the original Laplacian penalty with a pairwise Potts MRF and approximate inference.

3.4 RQ4: Can a context-aware prior be more robust than unconditional homophily?

Use edge weights based on feature similarity:

$$\text{Edge}(i, j) \wedge \text{Similar}(i, j) \Rightarrow \text{SameClass}(i, j).$$

3.5 Optional RQ5: What happens to predictive uncertainty?

Measure NLL/Brier score (and optionally calibration) to determine whether the prior improves probability quality or merely argmax accuracy.

4 Scope: What Is Essential vs. Optional

Tier	Include	Purpose
Must-have	MLP, GCN, GCN + Laplacian prior; Cora; low-label experiments; multiple seeds	Establish a clean baseline and reproduce the proposal faithfully
Strong core	Explicit Potts MRF + mean-field inference; synthetic SBM experiment	Makes the project deeply aligned with CMPT 727 and answers when the prior works
Strong extension	CiteSeer replication; similarity-weighted prior	Tests generalization and exception-aware logic
Stress test	Roman-empire heterophilous graph	Demonstrates what happens when homophily is a bad prior
Optional	Calibration, learned class-compatibility matrix, loopy BP comparison	Adds depth only if all core experiments are already stable

Important scope rule

Do not build five GNN architectures. The contribution should be the **probabilistic prior + controlled analysis**, not architecture benchmarking. One simple GCN is enough as the neural backbone.

5 Datasets to Use

5.1 Dataset A: Cora – primary dataset

Keep Cora because it is the dataset in the proposal and provides continuity. The proposal records 2,708 nodes, 1,433 features, 7 classes, and 5,278 undirected edges. Use it for all experiments.

Why: small, fast, standard citation network, and directly aligned with the homophily assumption.

5.2 Dataset B: CiteSeer – replication dataset

Use CiteSeer as the second real citation network. In the standard Planetoid formulation it contains 3,327 nodes, 3,703 features, 6 classes, and about 4,552 undirected links. Use exactly the representation loaded by PyTorch Geometric and compute/report the statistics from the loaded object rather than relying on copied numbers.

Why: tests whether the Cora finding transfers to a second citation graph without changing domain.

5.3 Dataset C: Synthetic stochastic block model – most important extra experiment

This is the experiment that makes the project research-like. Generate graphs where you control the label-homophily level while keeping expected degree approximately fixed.

Recommended setup:

- number of nodes: $n = 2800$;
- classes: $C = 7$ with roughly equal class sizes;
- expected degree: $k \approx 8$;

- target edge homophily:

$$h \in \{0.05, 0.15, 0.30, 0.50, 0.70, 0.90\};$$

- node features: class-conditioned Gaussian vectors, e.g.

$$x_i = \mu_{y_i} + \sigma_x \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, I).$$

For class size n_c and desired expected degree k , a convenient construction is

$$p_{\text{in}} = \frac{hk}{n_c - 1}, \quad p_{\text{out}} = \frac{(1-h)k}{n - n_c}.$$

This keeps the expected number of neighbors near k , while approximately controlling the same-class fraction of neighbors by h .

Critical control: choose the feature noise level σ_x once in a pilot so that features are informative but imperfect; then freeze it for the whole study. Do not retune it per homophily level.

5.4 Dataset D: Roman-empire – optional stress test

PyTorch Geometric provides the Roman-empire heterophilous dataset with 22,662 nodes, 32,927 edges, 300 features and 18 classes. Use it only after the Cora/CiteSeer/SBM pipeline works.

Why: an unconditional “neighbors should have the same label” prior should not automatically succeed on a heterophilous graph. If it hurts performance, that is a scientifically useful result.

5.5 Dataset decision

If time is limited, use:

Cora + CiteSeer + Synthetic SBM

If time is comfortable, add:

Roman-empire as a stress test

Do not add PubMed unless you specifically need a scalability experiment; the synthetic analysis is more valuable for the paper’s research question.

6 Project Setup from Zero

6.1 Step 0: Freeze the scientific contract

Before coding, write a one-page internal protocol containing:

- research questions H_1 – H_4 ;
- datasets;
- metrics;
- model list;
- seed list;
- hyperparameter search spaces;

- which validation metric selects hyperparameters;
- what is considered a final test evaluation.

The purpose is to prevent changing the experiment after seeing favorable test results.

6.2 Step 1: Create a reproducible repository

Recommended structure:

```
project/
  README.md
  requirements.txt
  configs/
    cora.yaml
    citeseer.yaml
    sbm.yaml
  src/
    data.py
    splits.py
    metrics.py
    models/
      mlp.py
      gcn.py
      laplacian_gcn.py
      potts_mrf.py
    inference/
      mean_field.py
    synthetic/
      sbm.py
    train.py
    evaluate.py
  scripts/
    run_baselines.sh
    run_low_label.sh
    run_lambda_sweep.sh
    run_sbm_grid.sh
  results/
    runs.csv
    summaries/
    predictions/
    checkpoints/
  figures/
  paper/
    main.tex
    references.bib
    appendix.tex
```

6.3 Step 2: Fix reproducibility

For every run save:

- dataset and split identifier;
- random seed;
- Git commit hash if using Git;
- full configuration;
- epoch selected by validation;
- validation and test metrics;
- model parameter count;
- training time if convenient.

Recommended seed set:

$$\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}.$$

Ten seeds are enough for a course project to expose instability without making the experiment enormous.

7 Model Progression

7.1 M0: Feature-only MLP

Use the node features but ignore the graph. This tells you how much information is available without relational structure.

$$p_i = \text{softmax}(f_\theta(x_i)).$$

This baseline is important: if the MLP already performs extremely well, the graph prior has less room to help.

7.2 M1: Standard GCN

Use a simple two-layer GCN. With $\tilde{A} = A + I$ and $\tilde{D}$ its degree matrix,

$$H^{(\ell+1)} = \sigma\left(\tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2} H^{(\ell)} W^{(\ell)}\right).$$

The supervised objective is cross-entropy on training nodes:

$$\mathcal{L}_{\text{CE}} = -\frac{1}{|\mathcal{T}|} \sum_{i \in \mathcal{T}} \log p_i(y_i).$$

Keep this architecture deliberately simple: e.g., two layers, hidden dimension 64, ReLU, dropout, Adam.

7.3 M2: GCN + Laplacian/logic regularization – proposal model

Your original regularizer is

$$\mathcal{L}_{\text{lap}} = \frac{1}{|E|} \sum_{(i,j) \in E} \|p_i - p_j\|_2^2.$$

Train with

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda \mathcal{L}_{\text{lap}}.$$

This can be related to graph Laplacian smoothing. If P stacks all node probability vectors and $L = D - A$,

$$\sum_{(i,j) \in E} \|p_i - p_j\|_2^2 \propto \text{Tr}(P^\top L P),$$

with the constant depending on the edge-counting convention.

7.4 M3: Probability-of-rule-violation regularizer

Under a factorized label approximation $q(Y) = \prod_i q_i(Y_i)$ with $q_i(c) = p_{ic}$,

$$\Pr(Y_i = Y_j) = \sum_c p_{ic} p_{jc} = p_i^\top p_j.$$

A probabilistically interpretable rule loss is therefore

$$\mathcal{L}_{\text{rule}} = \frac{1}{|E|} \sum_{(i,j) \in E} (1 - p_i^\top p_j).$$

This is a useful ablation because it expresses expected same-class rule satisfaction directly, rather than only matching probability vectors by squared distance.

7.5 M4: GCN unary potentials + explicit Potts MRF

Interpret the GCN logits $s_i(c)$ as unary evidence and define

$$p(\mathbf{y} \mid X, G) \propto \exp \left[\sum_i s_i(y_i) + \beta \sum_{(i,j) \in E} w_{ij} \mathbf{1}(y_i = y_j) \right].$$

For the basic Potts model, set $w_{ij} = 1$. This is an actual pairwise graphical model: the neural network supplies node-wise evidence and the MRF supplies relational evidence.

7.6 M4 inference: mean-field variational updates

Use

$$q(\mathbf{y}) = \prod_i q_i(y_i).$$

A standard coordinate form is

$$q_i(c) \propto \exp \left[s_i(c) + \beta \sum_{j \in N(i)} w_{ij} q_j(c) \right].$$

Normalize across classes after every update.

Implementation plan:

1. initialize $q_i = \text{softmax}(s_i)$;
2. perform $T \in \{1, 3, 5, 10\}$ synchronous mean-field iterations;
3. optionally clamp observed training labels to one-hot vectors during inference as a separate, clearly labeled experimental variant;
4. predict $\hat{y}_i = \arg \max_c q_i(c)$.

Fair-comparison rule

If you clamp training labels during MRF inference, report that as a separate collective-classification variant because it uses labeled nodes as explicit inference-time evidence. Do not silently compare it against a GCN that only uses labels during training.

7.7 M5: Similarity-weighted, exception-aware prior

Instead of treating every citation as equally trustworthy,

$$w_{ij} = g(\text{sim}(x_i, x_j)), \quad 0 \leq w_{ij} \leq 1.$$

A simple version is cosine similarity rescaled/clipped to $[0, 1]$.

Then

$$\text{Edge}(i, j) \wedge \text{Similar}(i, j) \Rightarrow \text{SameClass}(i, j).$$

This is the best stretch goal because it gives a principled explanation for cross-topic citations: the logic prior becomes soft and context-dependent.

8 Data Splits and Low-Label Protocol

8.1 Do not rely only on the public Planetoid split

The original public split is useful for comparability, but your central question is label scarcity. Create controlled training masks while keeping validation/test nodes fixed or using reproducible random partitions.

Recommended training label counts per class:

$$m \in \{1, 2, 5, 10, 20\} \text{ labels per class.}$$

For Cora this corresponds to 7, 14, 35, 70, 140 training labels.

For each m :

1. sample training nodes stratified by class;
2. keep validation and test sets disjoint;
3. use the same split for every model within a seed;
4. repeat across 10 seeds.

This creates paired comparisons: every model sees exactly the same data for seed s .

8.2 Validation discipline

Use validation data for:

- early stopping;
- selecting λ or β ;
- selecting mean-field iteration count T ;
- selecting similarity transformation parameters.

Never select these using test accuracy.

9 Hyperparameter Plan

Start small. Do not run a huge search.

Parameter	Suggested values
GCN hidden width	64 (optionally verify 32, 64)
Dropout	0.5
Learning rate	{0.01, 0.005}
Weight decay	$\{5 \times 10^{-4}, 10^{-3}\}$
Max epochs	300–500 with early stopping
Patience	50
Laplacian/rule strength λ	$\{0, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10\}$
MRF strength β	$\{0, 0.05, 0.1, 0.25, 0.5, 1, 2\}$
Mean-field iterations T	$\{1, 3, 5, 10\}$
Similarity weights	unweighted vs. cosine-weighted

Tune on validation data, preferably using a representative middle label rate (e.g., 5 labels/class), then verify sensitivity rather than re-optimizing everything for every condition.

10 Metrics to Compute and Save

10.1 Primary predictive metrics

Always save:

- test accuracy;
- macro-F1;
- validation accuracy and macro-F1;
- negative log-likelihood (NLL).

Optional:

- Brier score;
- expected calibration error (ECE);
- per-class F1.

10.2 Graph/prior diagnostics

Compute:

$$h_{\text{edge}} = \frac{\#\{(i, j) \in E : y_i = y_j\}}{|E|}.$$

Also compute adjusted homophily or label informativeness if feasible. Raw homophily alone can be misleading across graphs, so use these diagnostics to explain why a prior helps on one graph but not another.

10.3 Rule-satisfaction metric

For predicted distributions,

$$R = \frac{1}{|E|} \sum_{(i,j) \in E} p_i^\top p_j.$$

This lets you ask whether a model becomes more logically consistent and whether that consistency actually improves accuracy.

11 The Experiment Matrix

11.1 Experiment E0: Baseline sanity check

Goal: verify the pipeline before doing research experiments.

Run MLP and GCN on Cora with 20 labels/class. Confirm that:

- the loss decreases;
- validation accuracy is sensible;
- results are stable across a few seeds;
- no labels from validation/test enter training.

Do not proceed until this is clean.

11.2 Experiment E1: Reproduce the proposal

Compare:

GCN vs. GCN+Laplacian

on Cora for 20 labels/class, then sweep λ .

Save:

- validation curve for each λ ;
- final test mean $\pm$ std over seeds;
- rule satisfaction;
- training curves for one representative seed.

11.3 Experiment E2: Low-label study – core result

For

$$m \in \{1, 2, 5, 10, 20\},$$

compare MLP, GCN, GCN+Laplacian, and MRF variant.

Main plot:

Test accuracy vs. labels per class, with mean and error bars.

Also plot

$$\Delta \text{Acc}(m) = \text{Acc}_{\text{prior}}(m) - \text{Acc}_{\text{GCN}}(m).$$

This directly tests H_1 .

11.4 Experiment E3: Prior-strength sensitivity

For one low-label regime (e.g., 2 or 5 labels/class), plot accuracy against λ and/or β on a logarithmic x-axis.

Question:

Does performance show an intermediate optimum, with too much smoothing eventually hurting?

Also plot rule satisfaction versus prior strength. This can reveal a useful distinction: *more consistency is not necessarily more accuracy*.

11.5 Experiment E4: Explicit MRF vs. smoothness regularization

Compare:

- GCN;
- GCN + Laplacian regularizer;
- GCN + expected rule-violation regularizer;
- GCN unary logits + Potts MRF mean-field inference.

This answers whether the explicit probabilistic interpretation changes behavior or simply reproduces smoothing.

11.6 Experiment E5: Synthetic homophily sweep – centerpiece experiment

Run the synthetic SBM for each

$$h \in \{0.05, 0.15, 0.30, 0.50, 0.70, 0.90\}$$

and preferably at

$$m \in \{1, 5, 20\} \text{ labels/class.}$$

The most valuable figure is a heatmap with:

- x-axis: homophily h ;

- y-axis: labels per class;
- cell value: $\Delta\text{Acc} = \text{Acc}_{\text{MRF}} - \text{Acc}_{\text{GCN}}$.

Why this experiment matters

It can demonstrate that the same symbolic assumption is helpful when it matches the data-generating structure and harmful when it is violated. That is a stronger contribution than simply obtaining a higher Cora score.

11.7 Experiment E6: CiteSeer replication

Repeat only the most important experiments from Cora:

- 1, 5, 20 labels/class;
- GCN vs. best prior model;
- same evaluation procedure.

Do not duplicate every ablation. The purpose is external replication, not another full paper.

11.8 Experiment E7: Heterophily stress test – optional

Run GCN, unweighted prior, and similarity-weighted prior on Roman-empire.

Possible scientifically useful outcomes:

- unweighted homophily prior hurts;
- weighted prior hurts less;
- weighted prior becomes neutral or beneficial;
- both fail, demonstrating the limitation of the chosen symbolic rule.

Any of these can be a valid result if analyzed correctly.

11.9 Experiment E8: Uncertainty/calibration – optional

For Cora at low label rates, compare NLL/Brier score across models. If accuracy improves but NLL worsens, discuss overconfidence. If both improve, argue that the relational prior improves probability quality as well as classification.

12 What Results You Should Obtain vs. What You Should Never Chase

Do not decide the answer in advance

You should not aim for a particular accuracy improvement. The project is stronger if the experiments reveal a clear boundary between when the prior helps and when it fails. Save all runs, including negative ones.

What you **should obtain** means measurements and evidence:

- mean and standard deviation for each model/condition;
- paired improvement relative to GCN;
- low-label performance curves;
- prior-strength sensitivity curves;
- synthetic homophily heatmap;
- rule-satisfaction diagnostics;
- at least one cross-dataset replication;
- qualitative interpretation tied to your hypotheses.

Do not write claims such as “logic improves GNNs” unless the experiments support them. A more defensible conclusion may be “logic helps when the rule aligns with local graph structure and supervision is scarce.”

13 Run-Level Results File: Exact Schema

Maintain one machine-readable table such as `results/runs.csv` with one row per run:

```
dataset,split_seed,model,label_per_class,train_seed,
lambda,beta,mf_steps,weighted_prior,best_epoch,
val_acc,val_f1,test_acc,test_f1,test_nll,brier,
rule_score,train_seconds,checkpoint_path
```

This file should be the single source for all paper tables and plots. Do not manually type result values into plotting scripts.

13.1 Save predictions for important runs

For the best model of each main condition, save:

- logits;
- posterior probabilities;
- predicted labels;
- true labels;
- train/validation/test masks.

This makes error analysis and calibration plots possible later without retraining.

14 Statistical Reporting

For each condition report

mean $\pm$ standard deviation

over the same 10 seeds.

For model comparisons, compute the paired seed-wise difference

$$d_s = M_s - B_s,$$

where M_s is the prior model score and B_s is the GCN score using the same split/seed. Report the mean paired difference and optionally a 95% bootstrap confidence interval.

Do not over-emphasize a p -value in a small course project. Effect size, consistency across seeds, and a mechanism-aligned trend are more useful.

15 Figures and Tables to Produce

15.1 Main-paper figures

A strong final paper can be built around only four figures:

1. **Method diagram:** GCN unary evidence $\rightarrow$ relational prior $\rightarrow$ mean-field posterior.
2. **Low-label curve:** accuracy vs. labels/class on Cora, with confidence/error bars.
3. **Synthetic heatmap:** prior improvement vs. homophily and label rate.
4. **Prior sensitivity or stress test:** accuracy/rule satisfaction vs. λ/β , or Cora vs. Roman-empire.

15.2 Main-paper tables

Recommended:

- Table 1: dataset statistics and graph characteristics;
- Table 2: main low-label results (Cora/CiteSeer);
- Table 3: ablation of Laplacian vs. rule loss vs. MRF inference.

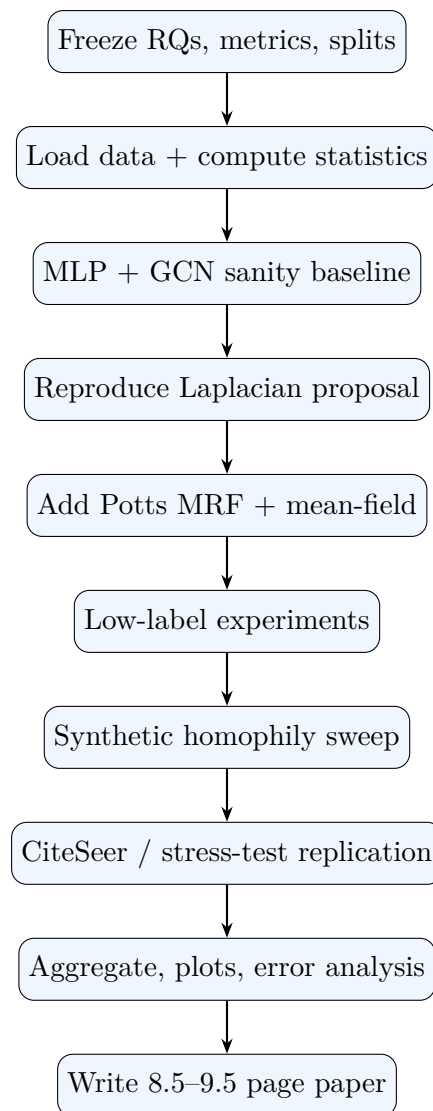
Put large hyperparameter grids and per-seed tables in the appendix.

16 Figure Design Rules

Every main figure should answer one question. Use the following checklist:

- axis labels include units/meaning;
- legend uses model names consistently across all figures;
- error bars state whether they are std, standard error, or CI;
- title/caption states the experiment and takeaway;
- no test-set tuning is implied;
- fonts remain readable at conference-paper size;
- figures are exported as vector PDF when possible.

17 End-to-End Workflow



18 A Practical 14-Day Schedule

Day	Focus	Exit condition
1	Protocol + repo	RQs, datasets, metrics, seeds, result schema fixed
2	Data pipeline	Cora/CiteSeer stats, masks, homophily diagnostics verified
3	MLP + GCN	Baselines train reproducibly; no leakage
4	Proposal model	Laplacian regularizer works; $\lambda = 0$ exactly recovers GCN objective
5	Sweep + debugging	Small λ sweep on one split; curves look stable
6	MRF formulation	Mean-field implementation passes unit tests on tiny graphs

Day	Focus	Exit condition
7	MRF on Cora	$\beta = 0$ recovers unary GCN predictions; inference stable
8	Low-label runs	Main Cora experiment launched/completed
9	Synthetic generator	SBM statistics match requested degree/homophily
10	Synthetic grid	Core homophily x label-rate study completed
11	CiteSeer	Replication completed
12	Optional extension	Weighted prior or Roman-empire stress test
13	Analysis	Final plots/tables generated only from <code>runs.csv</code>
14	Paper	Full conference-format draft assembled; appendix populated

If you have more time, spend it on repeated seeds, diagnostics, and writing quality before adding another model.

19 Implementation Checks Before Large Runs

19.1 GCN checks

- ☐ $\lambda = 0$ version equals baseline GCN implementation.
- ☐ train loss decreases.
- ☐ early stopping uses validation only.
- ☐ predictions have shape $N \times C$ and rows sum to one after softmax.

19.2 Laplacian loss checks

- ☐ zero if neighboring probability vectors are identical.
- ☐ nonnegative.
- ☐ each undirected edge counted once or normalization documented.

19.3 MRF checks

- ☐ $\beta = 0$ gives exactly the unary softmax posterior.
- ☐ every mean-field q_i sums to one.
- ☐ increasing β on a toy homophilous chain makes neighbor posteriors more similar.
- ☐ no test labels are clamped or used to form learned weights.

19.4 Synthetic graph checks

- ☐ empirical mean degree is near target k .
- ☐ empirical edge homophily tracks requested h .
- ☐ class proportions remain stable.

- $\square$ feature distribution is identical across graph-homophily settings except for random sampling.

20 How to Write the Final Conference Paper

The instructor will primarily evaluate the report, so treat the paper as the product and code as evidence supporting it.

20.1 Recommended 9-page budget

Section	Pages	What it must accomplish
Abstract	0.25	Problem, method, experiment, main finding in 150–200 words
1. Introduction	1.0–1.2	Motivation, precise gap, RQs, 3 contributions
2. Background / Related Work	0.8–1.0	GCNs, graph smoothness, MRFs/VI, homophily limitations
3. Methods	2.0–2.4	Baselines, Laplacian prior, Potts MRF, mean-field updates, weighted prior
4. Experimental Setup	0.8–1.0	Datasets, splits, metrics, seeds, hyperparameters
5. Results	2.0–2.4	Main tables/figures, low-label, synthetic sweep, ablation
6. Discussion / Future Work	0.8–1.0	Interpret mechanism, failure cases, limitations, future work
7. Conclusion	0.25–0.4	One clear take-home message

20.2 Introduction structure

Use five paragraphs:

1. Graph node classification is relational and labels are often scarce.
2. GNNs exploit structure, but the graph’s relational assumption is not always correct.
3. Your idea: represent homophily as an explicit prior, from smoothness regularization to an MRF.
4. The missing question: *when does that prior actually help?*
5. Contributions (three bullets in prose).

Suggested contribution claims should be descriptive, not inflated:

- a probabilistic formulation connecting GCN unary predictions to a pairwise label prior;
- a controlled study of label scarcity and graph homophily;
- empirical evidence characterizing when the prior helps, is neutral, or hurts.

20.3 Methods structure

A clean Methods section can be:

1. Problem setup and notation.

2. GCN baseline.
3. Laplacian regularization from the proposal.
4. Potts MRF formulation.
5. Mean-field approximate inference.
6. Similarity-weighted prior (if included).

Derive only the equations needed to understand the method. Put implementation pseudocode in the appendix unless it is central.

20.4 Results structure

Do not organize Results by “Experiment 1, Experiment 2” only. Organize by questions:

- **Relational priors help most in the low-label regime.**
- **Prior benefit depends strongly on graph homophily/informativeness.**
- **Explicit MRF inference and smoothness regularization have related but distinct behavior.**
- **Exception-aware weighting improves robustness** (only if supported).

Each subsection should contain: claim $\rightarrow$ evidence $\rightarrow$ interpretation $\rightarrow$ limitation.

20.5 Discussion/Future Work

Discuss at least:

- dependence on the correctness of the homophily prior;
- oversmoothing when λ/β is too large;
- transductive nature of Planetoid-style experiments;
- whether the synthetic graph captures real citation-network complexity;
- potential learned compatibility matrices or richer first-order rules;
- more sophisticated inference such as loopy BP or structured variational families.

21 Appendix Plan

Use the appendix for material that proves rigor without crowding the main paper:

- full hyperparameter table;
- exact split-generation algorithm;
- synthetic SBM generation details;
- extra seeds / per-seed results;
- additional λ/β curves;
- derivation of mean-field update;
- extra dataset results;

- implementation details and hardware/software versions.

The main 6–10 pages must remain understandable without the appendix.

22 Ablations That Matter

Prioritize these ablations:

1. **No prior vs. prior:** GCN vs. GCN+prior.
2. **Prior type:** squared probability smoothing vs. expected same-label rule vs. explicit MRF.
3. **Prior strength:** λ/β sweep.
4. **Information regime:** labels per class.
5. **Prior correctness:** synthetic homophily sweep.
6. **Edge reliability:** uniform vs. similarity-weighted prior.

Do not prioritize GAT vs. GraphSAGE vs. GIN unless the instructor specifically asks for architecture comparison.

23 Common Failure Modes and How to Avoid Them

Failure	Fix
Calling every smoothness penalty “logic”	Explicitly distinguish graph Laplacian regularization from the discrete MRF interpretation
Tuning on test accuracy	Freeze test set; use validation for all choices
One random split / one seed	Use paired repeated seeds and report variability
Too many datasets, shallow analysis	Prefer Cora + CiteSeer + controlled SBM over 8 benchmark datasets
Too many architectures	Keep GCN backbone fixed and study the probabilistic component
Only reporting accuracy	Add macro-F1 and NLL; add rule/graph diagnostics
Chasing SOTA	State that the objective is understanding prior behavior, not leaderboard optimization
Synthetic experiment changes degree and homophily together	Parameterize probabilities to hold expected degree approximately constant
Using test labels to compute edge weights	Any trainable/adaptive prior must avoid test-label information
Writing paper after all experiments	Write Methods and Experimental Protocol early; update Results later

24 Decision Rules When Experiments Disagree

- If Cora improves but CiteSeer does not: examine homophily, label informativeness, degree, and class-specific effects.

- If low-label improvements vanish at 20 labels/class: this supports the scarcity hypothesis rather than weakening the project.
- If large λ hurts: discuss over-smoothing / over-constraining the posterior.
- If the MRF and Laplacian methods are nearly identical: frame the result as evidence that the simpler relaxation captures much of the useful inductive bias.
- If the MRF is worse: inspect inference, calibration, and whether the pairwise prior is too rigid. A negative result is meaningful if the implementation is validated.
- If synthetic results show a transition as h increases: this can become the central paper result even if real-dataset gains are modest.

25 Minimum Viable Paper vs. Excellent Paper

25.1 Minimum viable, still solid

- Cora + CiteSeer;
- MLP, GCN, GCN+Laplacian;
- 1/2/5/10/20 labels per class;
- 10 seeds;
- λ sweep;
- strong report and interpretation.

25.2 Excellent version

Add:

- explicit Potts MRF;
- mean-field inference;
- controlled SBM homophily sweep;
- one context-weighted prior ablation;
- careful probability/uncertainty analysis.

25.3 Do not cross into unnecessary complexity

A learned 7×7 class-compatibility matrix,

$$\psi_{ij}(a, b) = \exp(\beta C_{ab}),$$

is a good future-work or optional extension, but should only be implemented if the excellent version above is already complete and documented.

26 Your First Working Session: Exact Checklist

Start here

1. Create the repository/folder structure.
2. Load Cora using PyTorch Geometric.
3. Print and save: nodes, edges, features, classes, class counts, degree statistics, edge homophily.
4. Generate one stratified split with 20 labels/class and save the mask.
5. Implement MLP and two-layer GCN.
6. Train each for seeds 0, 1, 2 only.
7. Verify test labels never influence training/selection.
8. Create `runs.csv` immediately, even before the sophisticated models exist.
9. Only after this is reproducible, add the Laplacian loss.

At the end of the first session you should not have a novel result. You should have a trustworthy experimental foundation.

27 Final Submission Checklist

27.1 Scientific completeness

- ☐ Research questions are explicit and answered.
- ☐ Every main claim has a table/figure supporting it.
- ☐ Baselines use identical splits.
- ☐ Hyperparameters are selected using validation, not test.
- ☐ Variability across seeds is reported.
- ☐ Negative/failure cases are discussed.

27.2 Report quality

- ☐ Main paper is 6–10 pages excluding references.
- ☐ Abstract states actual findings, not intentions.
- ☐ Introduction contains a clear gap and contributions.
- ☐ Methods are reproducible from the paper + appendix.
- ☐ Captions are self-contained.
- ☐ Discussion distinguishes evidence from speculation.
- ☐ Appendix contains extra detail but is not required to understand the core argument.

27.3 Reproducibility

- ☐ Seeds, software versions, and hardware noted.
- ☐ Dataset source and exact split protocol stated.
- ☐ Hyperparameter search ranges reported.
- ☐ Results can be regenerated from `runs.csv` and plotting scripts.
- ☐ No copied numbers or plots without traceable experiment files.

28 Recommended Reading: Only What Directly Helps the Project

Read these selectively rather than trying to survey all graph learning:

1. Kipf and Welling (2017): GCN baseline and semi-supervised node classification.
2. Platonov et al. (2023): adjusted homophily, label informativeness, and why raw homophily can be misleading.
3. Platonov et al. (2023): heterophilous graph benchmark and evaluation pitfalls.
4. Murphy, *Probabilistic Machine Learning: Advanced Topics*: MRFs and variational inference chapters already used in CMPT 727.
5. Optional recent perspective: neuro-symbolic probabilistic reasoning on graph data, to position the project without claiming novelty beyond what you actually implement.

29 The One-Sentence Standard for the Final Paper

Before submitting, you should be able to complete this sentence with evidence:

Final story

A relational prior is most useful when _____, becomes harmful when _____, and the controlled experiments show this because _____.

If the final Results section supports a sentence of that form, the project will feel like a coherent research study rather than a collection of experiments.

References and Official Resources

References

- [1] W. Ahmed. *Node Classification on Citation Networks using Graph Neural Networks with Logic Regularization*. CMPT 727 Project Proposal, Spring 2026.
- [2] T. N. Kipf and M. Welling. Semi-Supervised Classification with Graph Convolutional Networks. International Conference on Learning Representations (ICLR), 2017. <https://openreview.net/forum?id=SJU4ayYgl>
- [3] Z. Yang, W. W. Cohen, and R. Salakhutdinov. Revisiting Semi-Supervised Learning with Graph Embeddings. ICML, 2016. <https://arxiv.org/abs/1603.08861>
- [4] PyTorch Geometric. Planetoid dataset documentation (Cora, CiteSeer, PubMed). https://pytorch-geometric.readthedocs.io/en/2.8.0/generated/torch_geometric.datasets.Planetoid.html
- [5] O. Platonov, D. Kuznedelev, A. Babenko, and L. Prokhorenkova. Characterizing Graph Datasets for Node Classification: Homophily-Heterophily Dichotomy and Beyond. NeurIPS, 2023. <https://arxiv.org/abs/2209.06177>
- [6] O. Platonov, D. Kuznedelev, M. Diskin, A. Babenko, and L. Prokhorenkova. A Critical Look at the Evaluation of GNNs under Heterophily: Are We Really Making Progress? ICLR, 2023. <https://arxiv.org/abs/2302.11640>
- [7] PyTorch Geometric. HeterophilousGraphDataset documentation. https://pytorch-geometric.readthedocs.io/en/2.5.1/generated/torch_geometric.datasets.HeterophilousGraphDataset.html
- [8] ICLR. ICLR 2026 Author Guide and official template. <https://iclr.cc/Conferences/2026/AuthorGuide>
- [9] NeurIPS. NeurIPS 2026 Main Track Handbook. <https://neurips.cc/Conferences/2026/MainTrackHandbook>